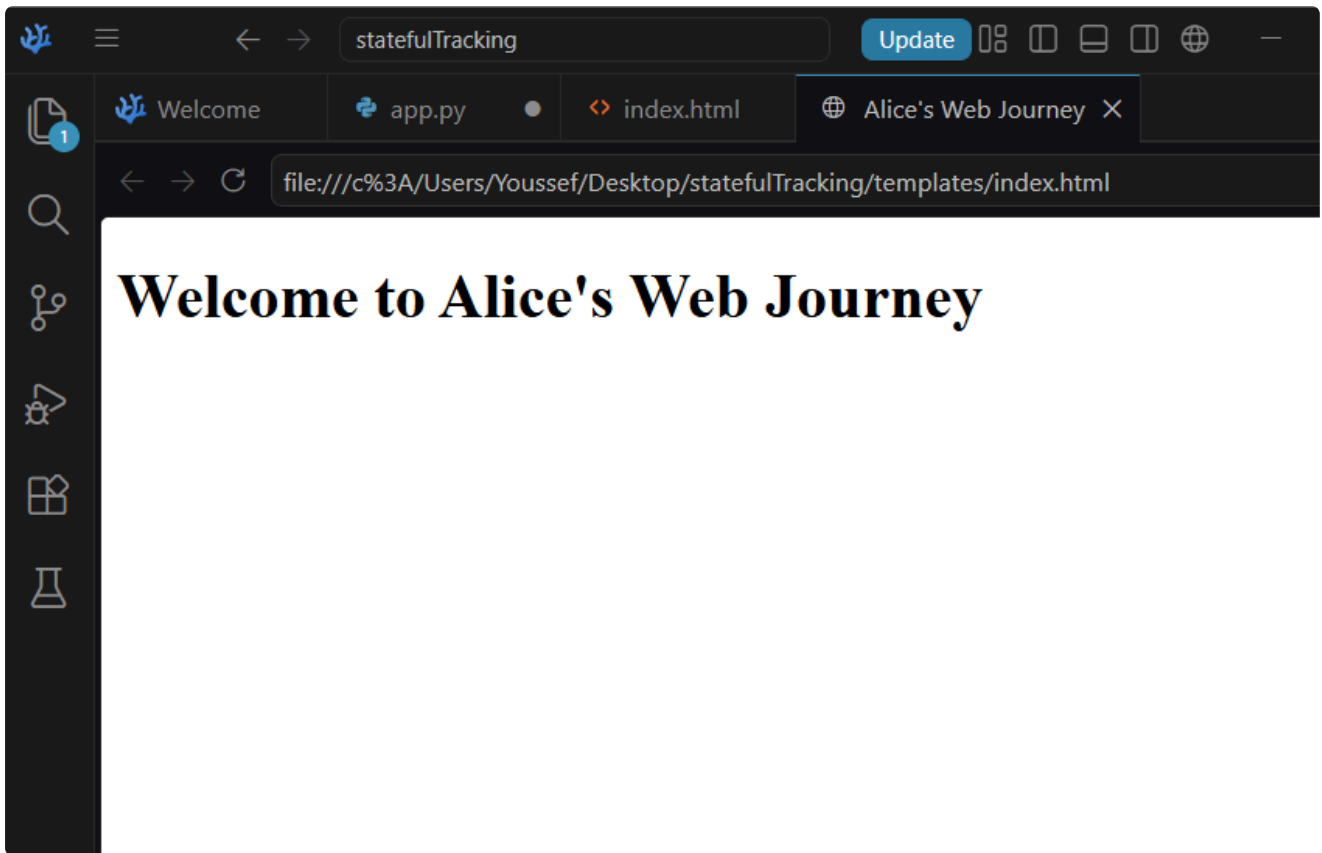
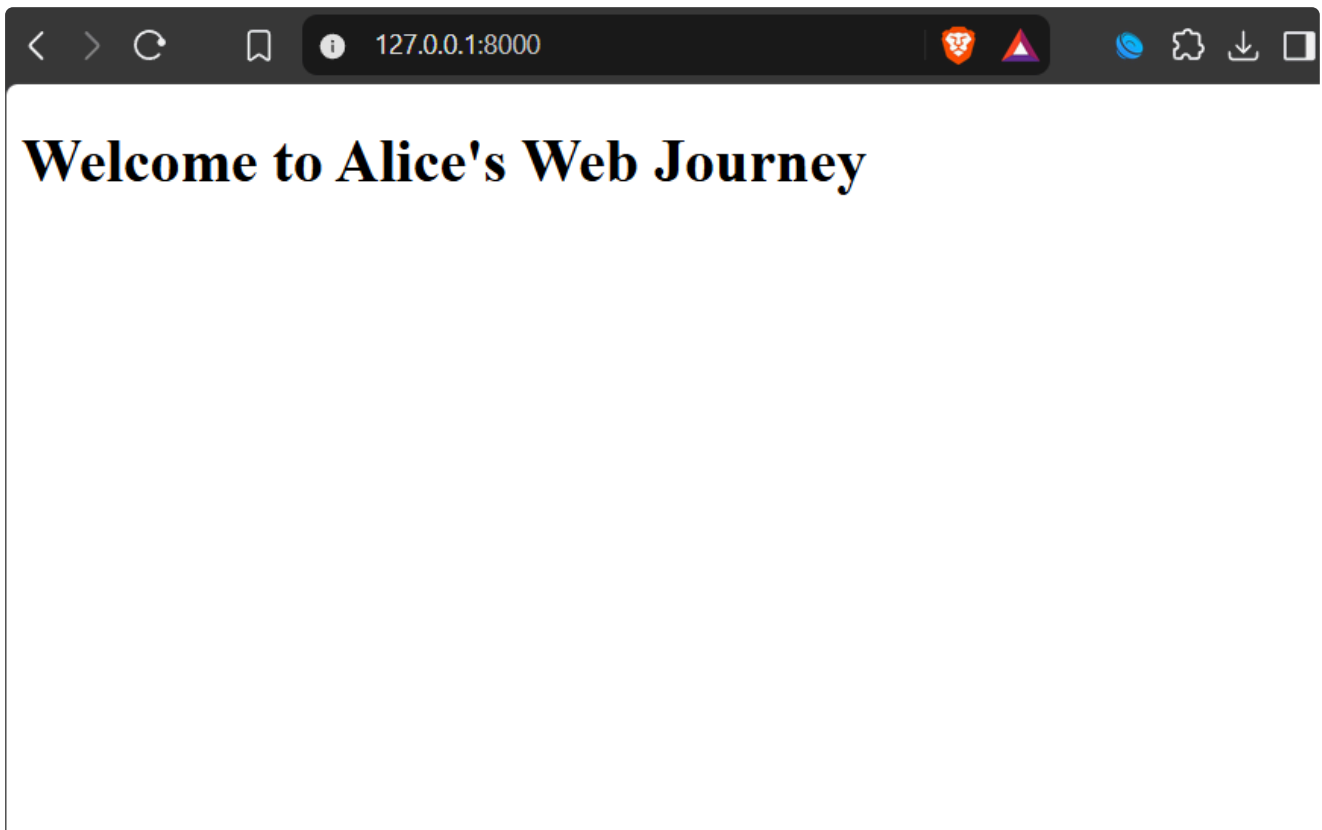


The html code rendered:

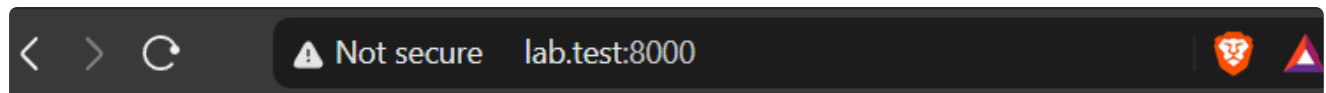


The flask application running on port 8000:



host name edited:

```
# localhost name resolution is handled within DNS itself.
#       127.0.0.1       localhost
#       ::1            localhost
127.0.0.1 lab.test
```



Welcome to Alice's Web Journey

we can read request/response headers:

```
(venv) PS C:\Users\Youssef\Desktop\statefulTracking> & c:/Users/Youssef/Desktop/statefulTracking/venv/Scripts/python.exe c:/Users/Youssef/Desktop/statefulTracking/app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:8000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 118-327-491

--- HTTP REQUEST HEADERS ---
Host: lab.test:8000
Connection: keep-alive
Cache-Control: max-age=0
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8
Sec-Gpc: 1
Accept-Encoding: gzip, deflate
Accept-Language: en-US,en;q=0.5
```

```
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng
*/*;q=0.8
Sec-Gpc: 1
Accept-Encoding: gzip, deflate
Accept-Language: en-US,en;q=0.5
```

---HTTP RESPONSE HEADERS ---

```
Content-Type: text/html; charset=utf-8
Content-Length: 208
X-Lab-Message: Hello from the Flask server
```

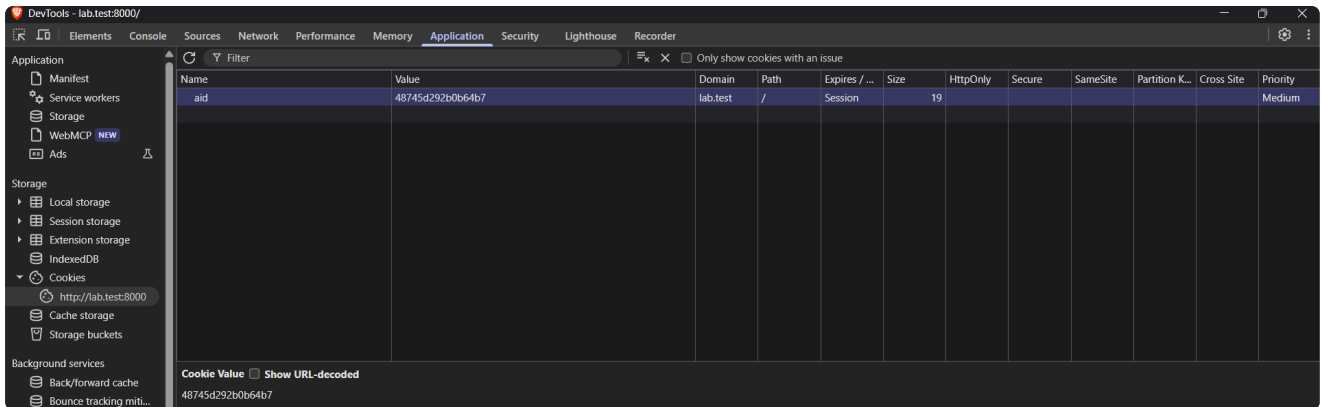
```
127.0.0.1 - - [22/Sep/2026 11:36:58] "GET / HTTP/1.1" 200 -
[]
```

from browser dev tool:

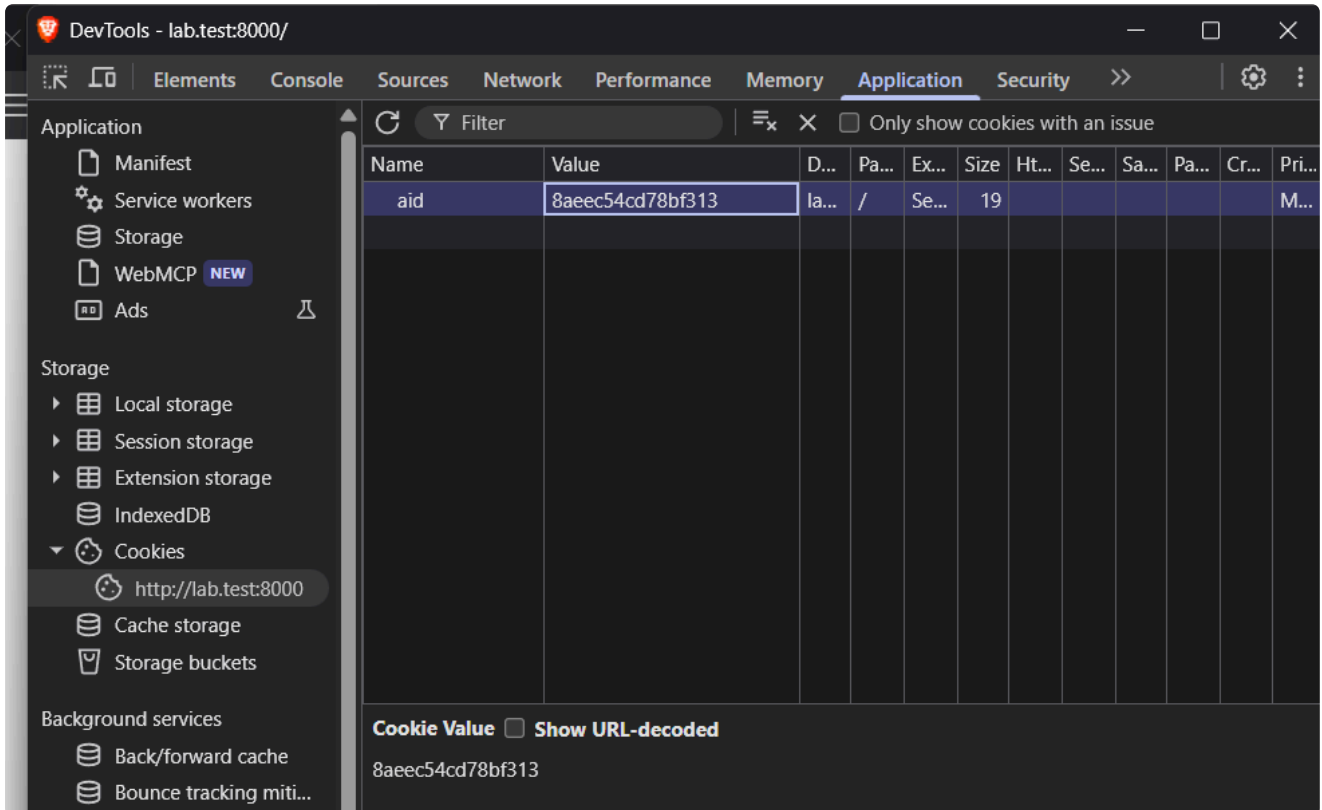
name	Headers	Preview	Response	Initiator	Timing	Adblock
lab.test	Remote Address		127.0.0.1:8000			
	Referrer Policy		strict-origin-when-cross-origin			
	▼ Response headers		☐ Raw			
	Connection		close			
	Content-Length		208			
	Content-Type		text/html; charset=utf-8			
	Date		Tue, 22 Sep 2026 11:39:01 GMT			
	Server		Werkzeug/3.1.8 Python/3.14.6			
	X-Lab-Message		Hello from the Flask server			
	▼ Request headers		☐ Raw			

▼ Request headers	☐ Raw
Accept	text/html,application/xhtml+xml,application/xml;q=0.9, image/avif,image/webp,image/apng,*/*;q=0.8
Accept-Encoding	gzip, deflate
Accept-Language	en-US,en;q=0.5
Cache-Control	max-age=0
Connection	keep-alive
Host	lab.test:8000
Sec-Gpc	1
Upgrade-Insecure-Requests	1
User-Agent	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36

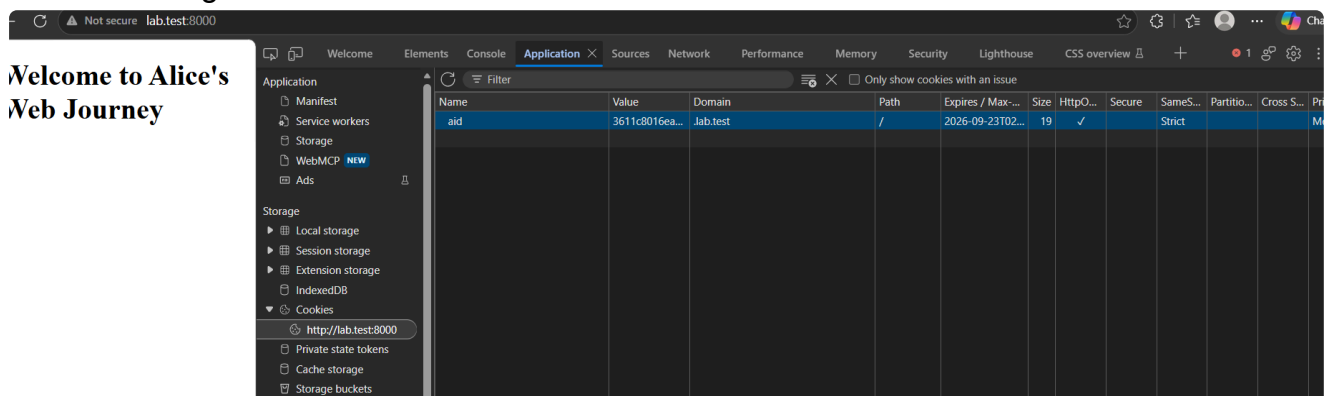
after adding cookies and checking the browser dev tools:



after restarting the browser, the cookies are still there, but with different identifier

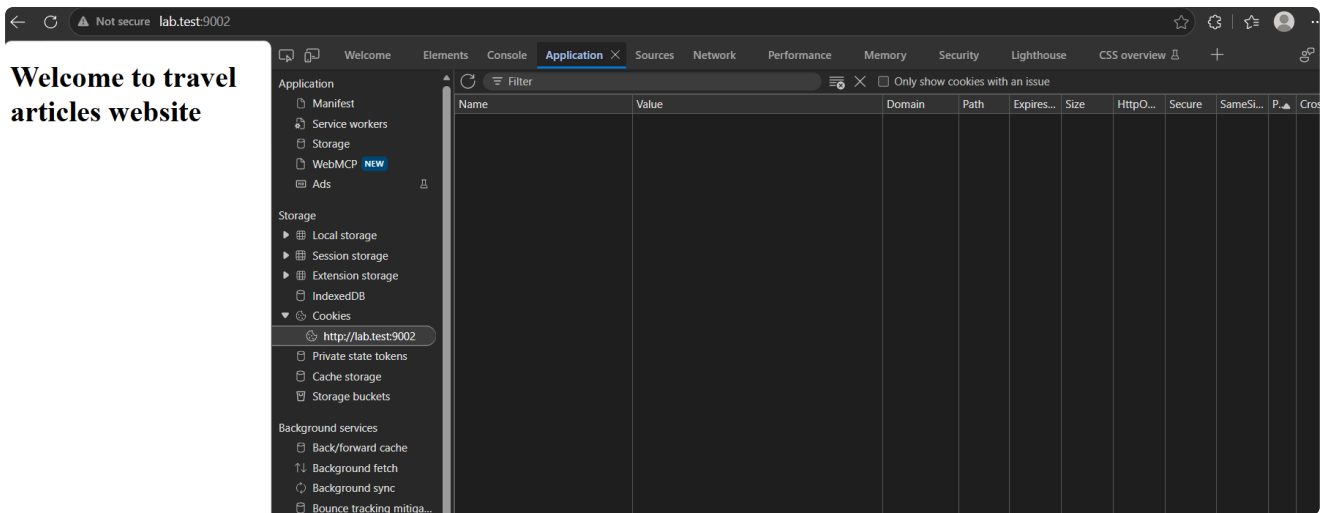


setting max_age to 5000 seconds, samesite to strict, and path to "/" domain to "lab.test" leads to those changes in the cookies:

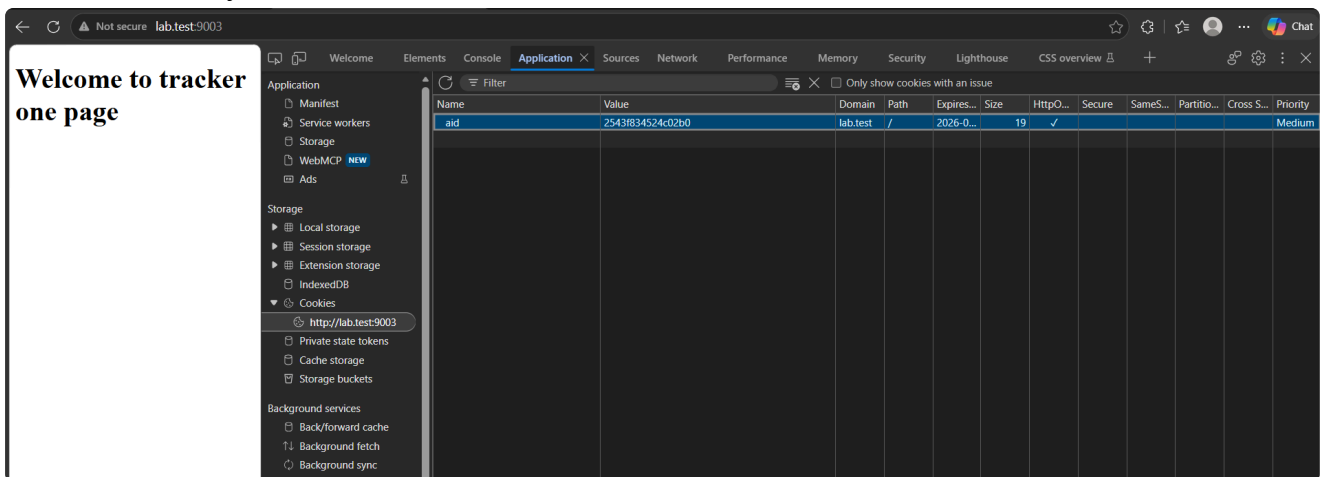


Challenge 1

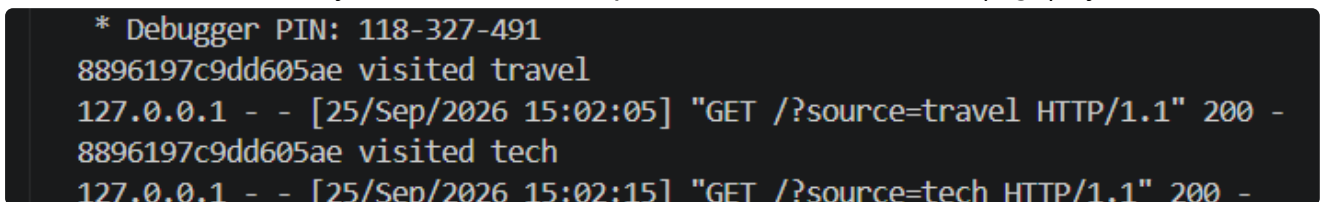
Embedding tracker one to the publishers makes publishers have no cookies on them as shown here:



cookies can only be found on the tracker one:

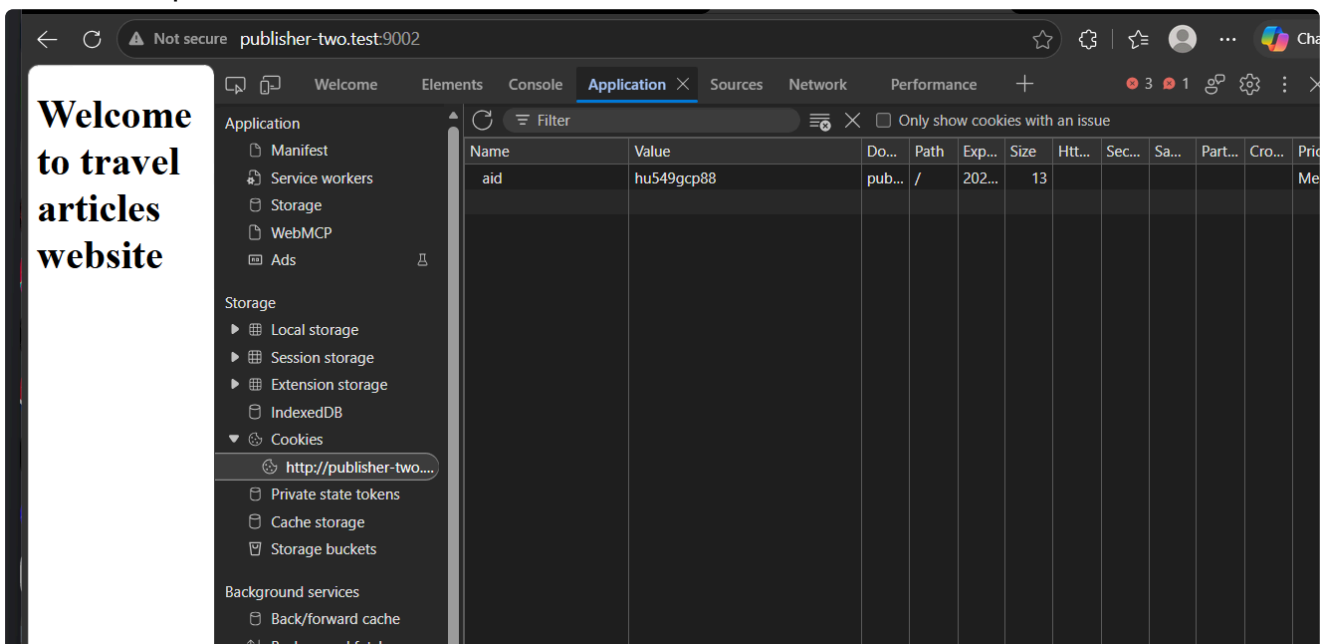


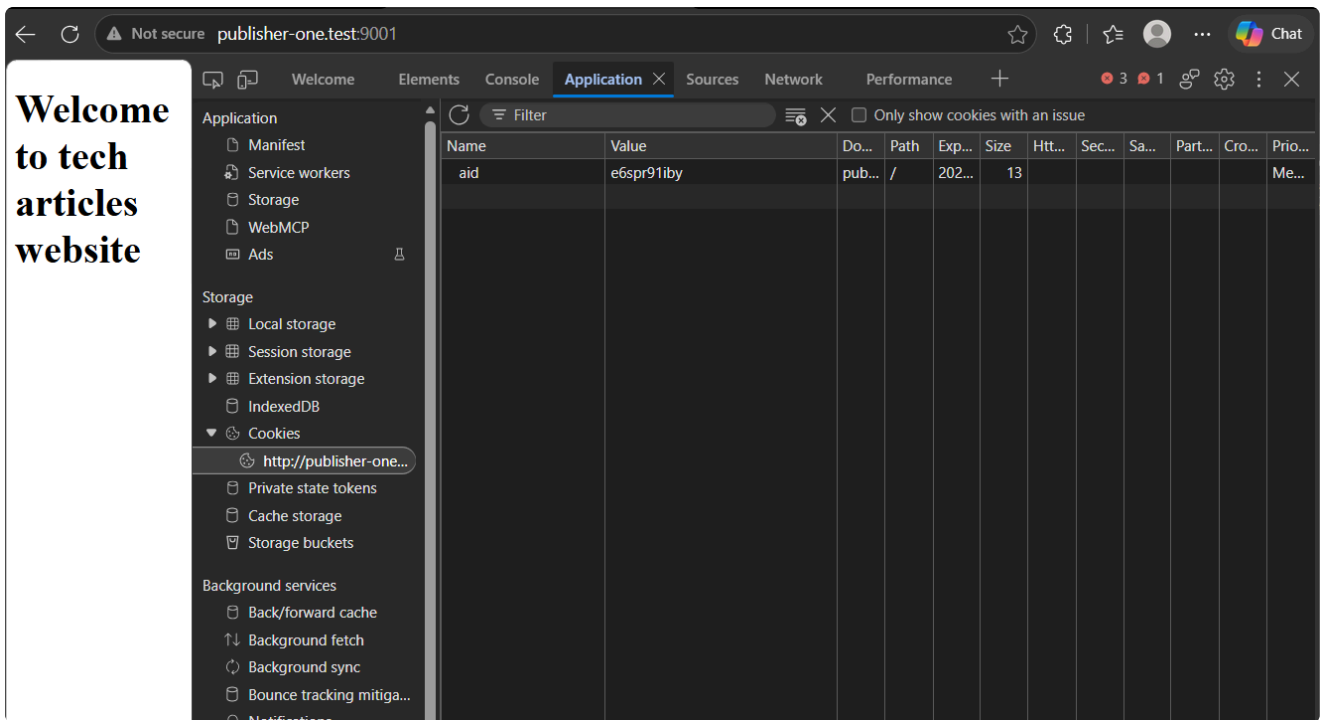
now the tracker can fully determine and keep track of websites visited(logs) by that browser:



Challenge 2

each of the publishers has its own cookie id:





and we get a clear view as analytics on the list of visits for each website:

```
{
  "e6spr91iby": [
    "Tech articles"
  ],
  "hu549gcp88": [
    "Travel atricles"
  ]
}
```

For the questions:

The two identifiers are different ,and thats because the analytics tracker works independently in each website generating "a unique id for that specific browser on that website"

no the analytics server does not receive them automatically because browsers only send cookies that match the target domain of the request since the first party cookies belong to the publisher websites the browser does not attach them when sending requests to the analytics server so the javascript script had to read the cookie manually and send the identifier inside the url parameters instead

Bonus cookie syncing

Tracker one assings this id to the browser and sends that to tracker 2 when publisher 1 is visited by our browser(tracker 2 does the same when that browser visits publisher 2)



and here is the linking on action:



Tracker Two ID (Internal)	Tracker One ID (Partner)
50c8d04a	0dbaad71
62970df4	9fd6ae3f